LinkedIn edunet foundation

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

PRESENTATION 2026

Smart Faculty Workload Management & Time Allocation System

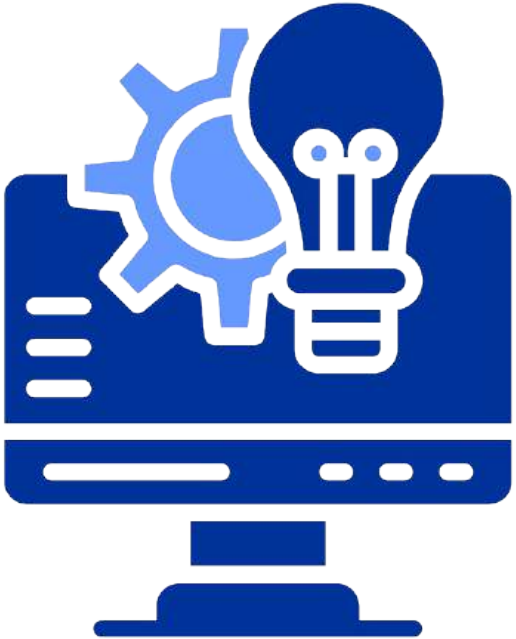
Mentor Name
Ms.UmaMaheshwari R

Dr Manisha Khanna

Student Email
khannamanishamgmt@gmail.com

Outline

- Abstract of the Project
- Problem Statement
- Proposed Solution
- System Architecture
- Embedded Video of Project
- Conclusion
- Future Scope



Abstract

At the beginning, the data is collected in Microsoft Excel. It includes workload details of around 50 faculty members from different departments. The dataset covers various activities such as teaching, research, administrative work, mentoring, and examination duties. This helps in capturing a complete picture of faculty responsibilities.

Once the data is collected, the next step is cleaning and preparing it. Missing values are checked, formats are standardized, and all entries are verified to ensure accuracy. This makes sure that further analysis is reliable.

After cleaning, the data is analyzed using Excel. Different formulas are used to calculate total workload, workload percentage, and to identify whether a faculty member is overloaded or balanced. Pivot tables are especially useful here as they help in summarizing the data and identifying patterns across departments.

The processed data is then moved to Power BI, where it is transformed into an interactive dashboard

Problem Statement

Managing faculty workload in higher education institutions is complex due to multiple responsibilities such as teaching, research, mentoring, and administrative tasks. These activities are often recorded in separate systems or spreadsheets, leading to:

Lack of transparency in workload distribution

Difficulty in tracking total workload

Manual errors and inconsistencies

Underrepresentation of actual faculty efforts

This affects faculty satisfaction and decision-making by administrators.

Proposed Solution

The scope of this project is focused on analyzing and improving faculty workload management using data analytics and visualization tools. It involves the creation of a structured dataset representing various faculty responsibilities and applying analytical techniques to evaluate workload distribution.The project is limited to a simulated dataset of faculty members and does not involve real-time data integration from institutional systems. The analysis is carried out using tools such as Microsoft Excel and Power BI, along with AI support from Microsoft Copilot for generating insights.The system is designed to provide a clear understanding of workload patterns, identify imbalances, and suggest possible improvements. However, it does not include automated workload allocation or direct integration with institutional databases.Overall, the project serves as a practical model that demonstrates how data-driven methods and AI tools can be used to enhance decision-making in academic workload management.

System Architecture

Data Input / Source

The system uses a structured dataset developed in Microsoft Excel, containing workload information for approximately 50 faculty members across multiple departments. The dataset includes various workload components such as teaching hours, research activities, administrative tasks, mentoring duties, examination responsibilities, evaluation work, and meeting hours. This dataset is designed to reflect realistic academic workload scenarios.

Processing / Analysis

The cleaned dataset is analyzed using Excel and Power BI. In Excel, formulas are applied to compute key metrics such as total workload, percentage workload, and workload classification (e.g., balanced or overloaded). Pivot tables help summarize and identify trends across departments and faculty members.

AI / Tool Integration

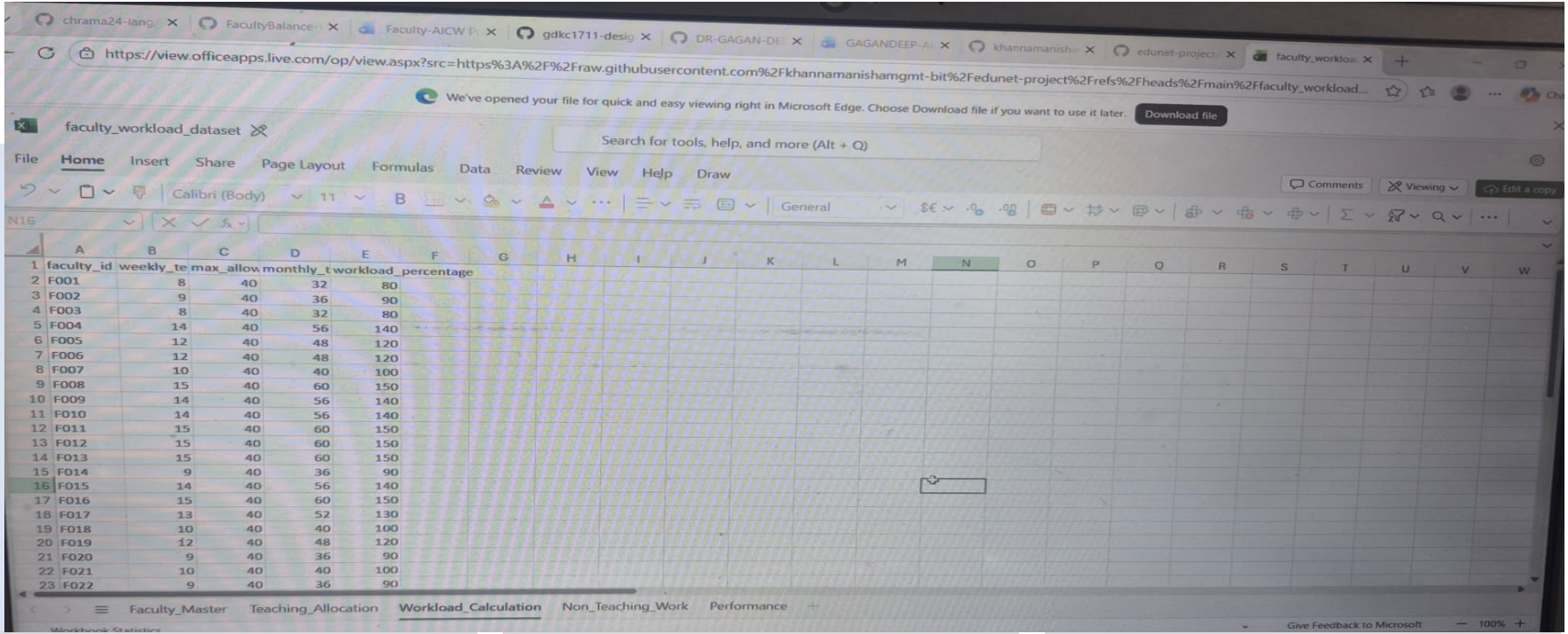
Power BI is then used to build a structured data model, enabling deeper analysis through measures and aggregations. Additionally, Microsoft Copilot assists in generating insights, identifying anomalies, and suggesting workload optimization strategies.

Output / Dashboard

The Power BI dashboard titled **"Smart Faculty Workload Management"** is designed to provide a comprehensive, interactive, and user-friendly representation of faculty workload distribution across various academic and non-academic activities. It enables stakeholders such as administrators, HR personnel, and faculty members to understand workload patterns clearly and make informed decisions.

Live Demo of Project

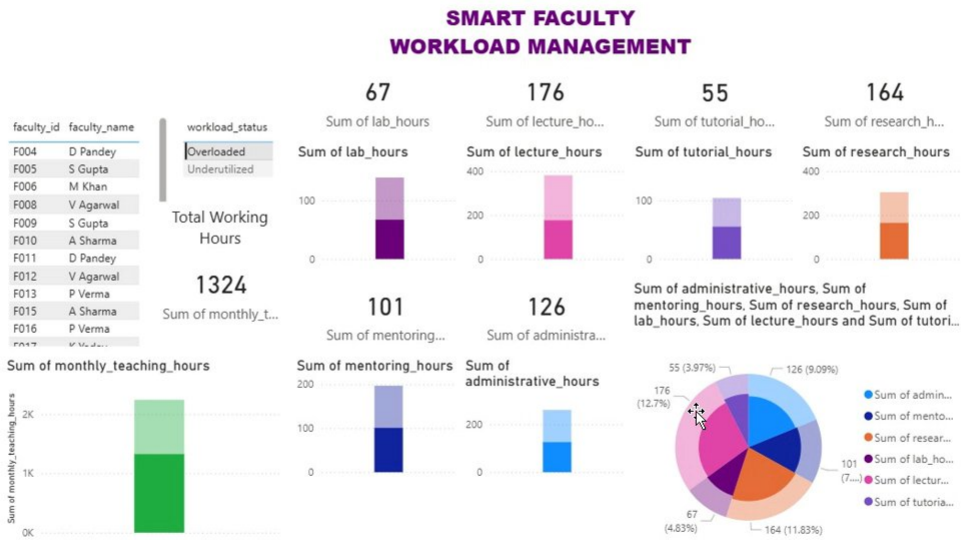
Here's the list of some demo screenshots / features of the project:



faculty_id	weekly_te	max_allow	monthly_t	workload_percentage
F001	8	40	32	80
F002	9	40	36	90
F003	8	40	32	80
F004	14	40	56	140
F005	12	40	48	120
F006	12	40	48	120
F007	10	40	40	100
F008	15	40	60	150
F009	14	40	56	140
F010	14	40	56	140
F011	15	40	60	150
F012	15	40	60	150
F013	15	40	60	150
F014	9	40	36	90
F015	14	40	56	140
F016	15	40	60	150
F017	13	40	52	130
F018	10	40	40	100
F019	12	40	48	120
F020	9	40	36	90
F021	10	40	40	100
F022	9	40	36	90

AICW Project Report final - Word
Manisha Khanna
File Home Insert Draw Design Layout References Mailings Review View Help Tell me what you want to do
Share
Paste Cut Copy Format Painter
Clipboard
Arial 11
Font
Paragraph
¶ heading... ¶ heading... ¶ heading... ¶ List Para... ¶ Normal ¶ Strong1 ¶ No Spac... ¶ Heading 1
Styles
Find Replace Select
Editing
Add-ins

distribution across various academic and non-academic activities. It enables stakeholders such as administrators, HR personnel, and faculty members to understand workload patterns clearly and make informed decisions.



Total Metrics Overview (KPI Cards)

The dashboard displays key performance indicators (KPIs) at the top, summarizing the total workload across different activity categories.

4215 words

Conclusion

This project successfully demonstrates the application of data analytics and AI tools in solving real-world problems in academic institutions. By integrating Excel, Power BI, and Microsoft Copilot, the project provides a structured approach to managing faculty workload.

The interactive dashboard enables stakeholders to visualize workload distribution clearly and identify imbalances. The use of AI tools further enhances the efficiency of analysis by generating quick insights and summaries.

Future Scope

All project objectives were achieved, including data cleaning, analysis, visualization, and AI integration. The system developed is scalable and can be extended to include real-time data and predictive analytics in the future. Overall, the project contributes to improving efficiency, transparency, and fairness in faculty workload management and demonstrates the practical value of AI and business intelligence tools in the education sector.

Thank You

AICW Fellowship | Student Project Presentation 2026

LinkedIn | Edunet Foundation